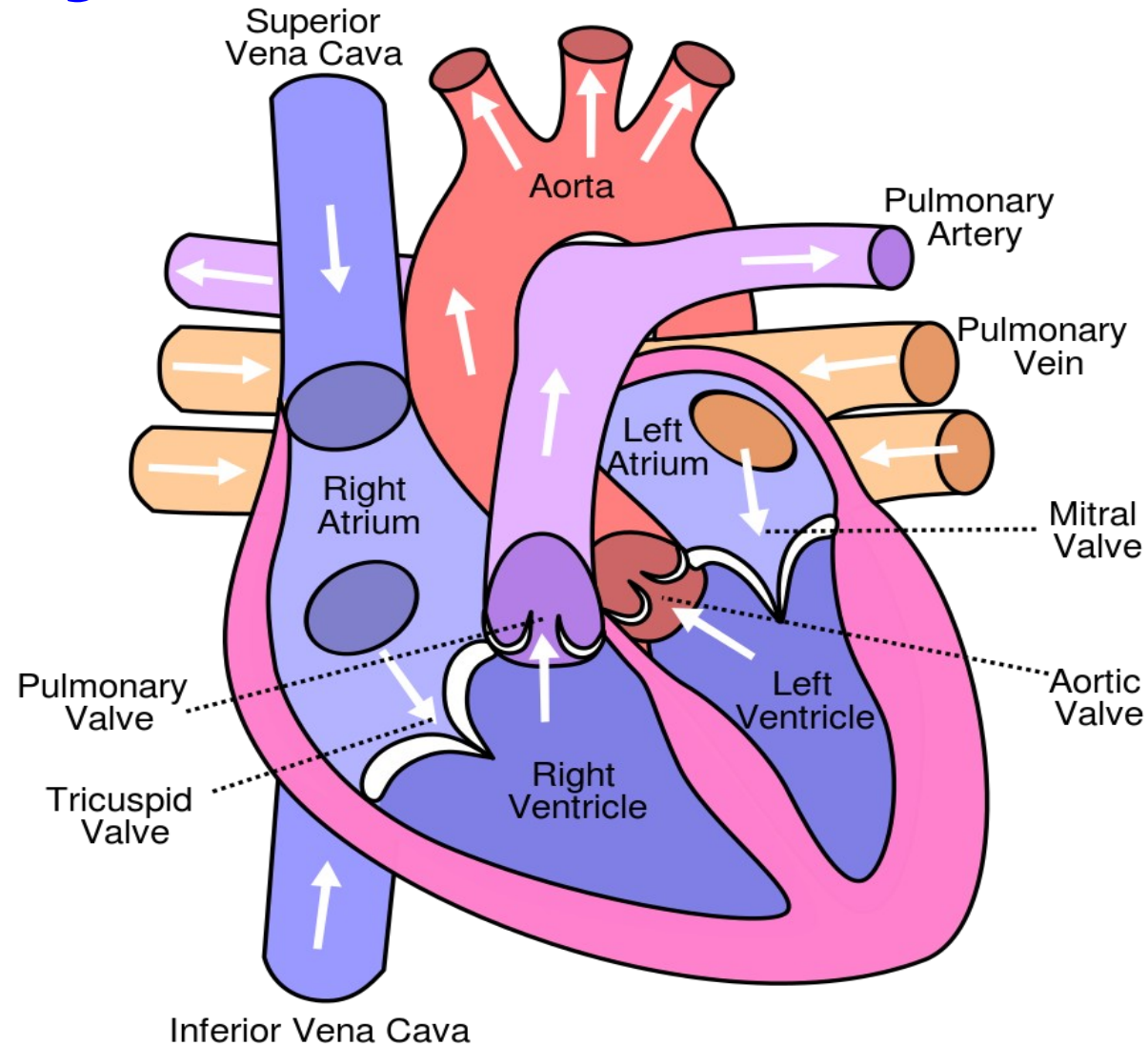


Pharmacotherapy of VHD

Anatomy of the valves



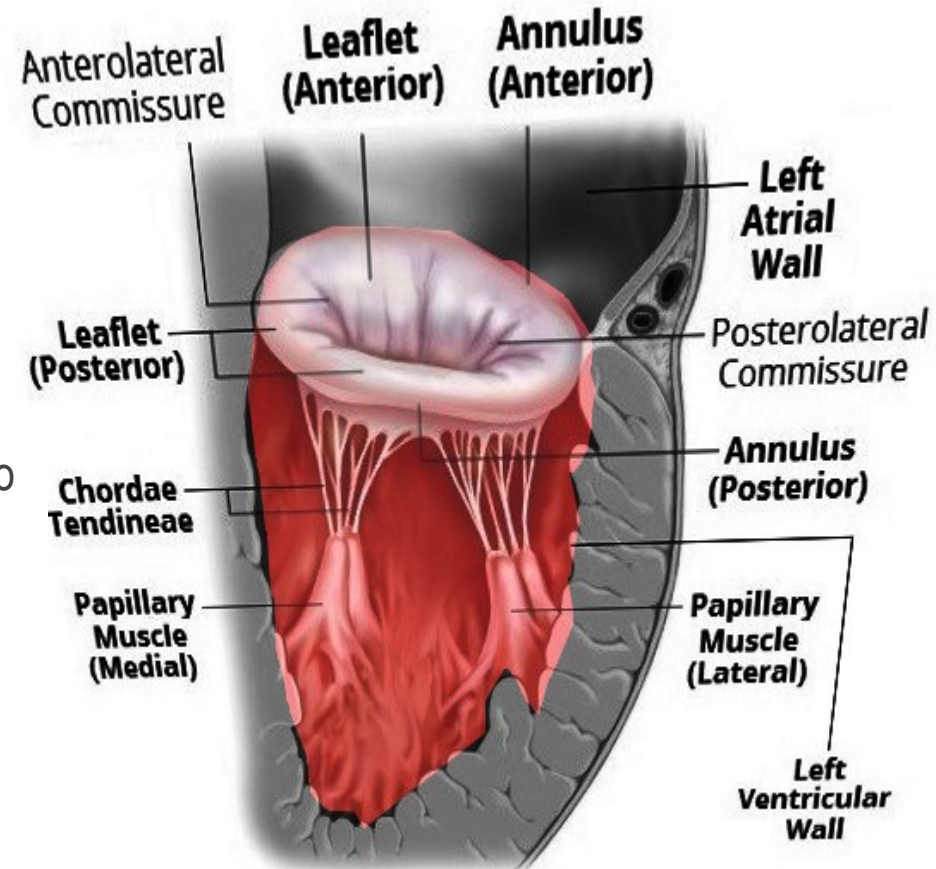
Valves of heart

Tricuspid and mitral valves

- A-V valves
- Prevent backflow of blood to the atria during systole

Attach mitral leaflets to papillary muscle tips

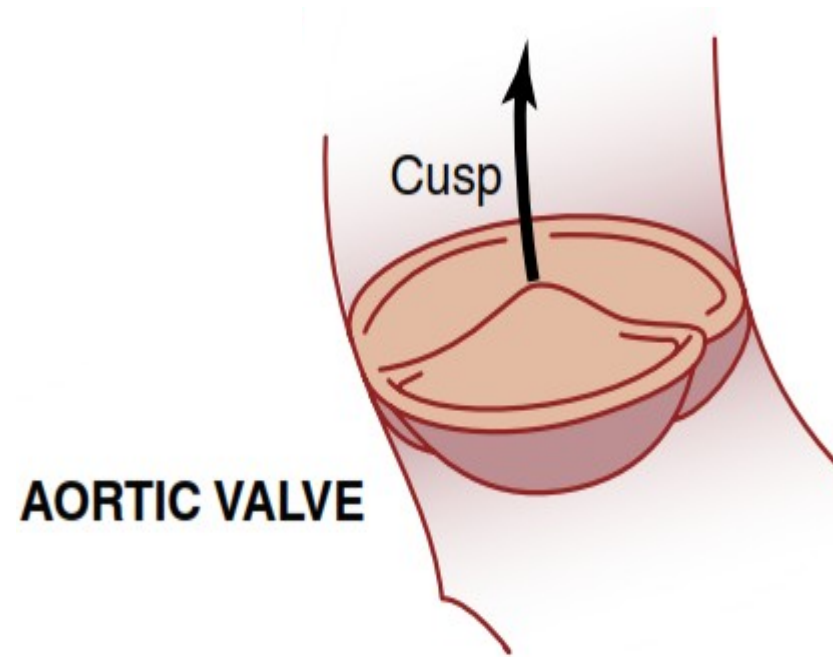
As a sphincter; completely closes the leaflets



Valves of heart

Aortic and pulmonic

- Prevent backflow of blood during diastole
- 1. Snap closing
- 2. High velocity
- 3. Supported by strong, yet very pliable, fibrous tissue



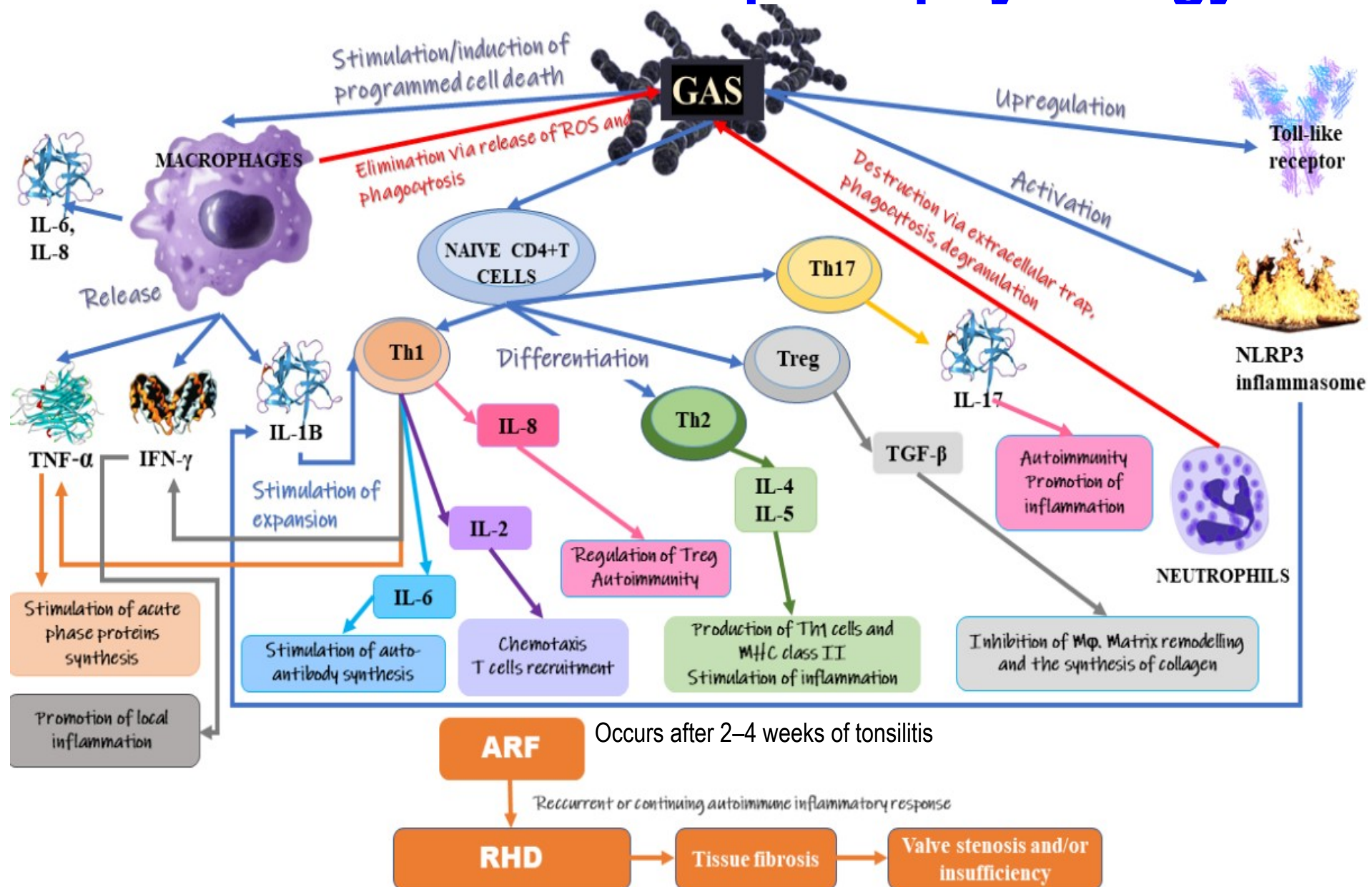
Valvular Heart Disease (VHD)

- Any deviation from normal valve anatomy or function
- **Primary VHD;**
 - Ranks well below CAD stroke, hypertension, obesity, and diabetes
- **Rheumatic fever;**
 - The dominant cause in LMICs
 - ✓ Group A streptococcal pharyngitis
- Other causes could be possible
- The lesions associated with the four valves;
 1. Stenosis
 2. Regurgitation

Etiology of VHD

- Can present acutely (e.g., IE) or progress slowly over time (e.g., calcific AS)
- Causes of VHD are multifactorial but have predominant etiologies
 - ✓ Aortic Stenosis (AS) – by valvular calcification
 - ✓ Aortic regurgitation (AR) - abnormal valve leaflets (e.g., IE, congenital bicuspid valve) or aortic root disease (e.g., Marfan aortic dissection)
 - ✓ Mitral stenosis (MS) - Rheumatic fever
 - ✓ Acute mitral regurgitation (MR) - chordae tendineae or papillary muscle rupture (e.g., from acute myocardial infarction)
 - ✓ Chronic MR - LV dilation and remodeling.
 - ✓ Tricuspid Stenosis (TS)- Rheumatic fever
 - ✓ Pulmonic stenosis (PS)- Congenital

Acute Rheumatic fever: pathophysiology



Diagnosis of VHD

- Physical examination
- ECG and chest radiography - to assess rhythm and pulmonary congestion or other lung pathology
- Echocardiography (using TTE) is the gold standard to confirm and in assessing severity and prognosis
- Cardiac catheterization - to assess pressures in the cardiac chambers
- Exercise stress testing - to resolve discrepancies between clinical signs/symptoms and other objective testing
- Cardiac MRI
- Computed tomography

Staging & progression of VHD

Stage	Definition	Description
A	At risk	Risk factors for VHD
B	Progression	Progressive VHD of mild to moderate severity without symptoms
C	Asymptomatic severe	Asymptomatic with criteria for severe VHD: C1: Compensated left or right ventricle in asymptomatic severe VHD C2: Decompensated left or right ventricle in asymptomatic severe VHD
D	Symptomatic severe	Symptomatic VHD

Sign & symptoms of VHD

Valve Disorder	Definition	Stage D Symptoms
AS	Narrowing of the aortic valve with reduced flow through the valve	DOE, exercise intolerance, angina, syncope or presyncope, HF
AR	Valvular dysfunction that allows flow of blood back into the LV	DOE, angina, or severe HF symptoms
MS	Narrowing of the mitral valve with reduced flow through the valve	Decreased exercise tolerance, DOE
Acute MR	Acute valvular dysfunction that allows flow of blood back into the left atrium and pulmonary veins during systole	Decreased exercise tolerance, DOE
Chronic MR	Gradual valvular dysfunction that allows flow of blood back into the left atrium and pulmonary veins during systole	HF symptoms that persist with revascularization and medical therapy, decreased exercise tolerance, DOE
TR	Valvular dysfunction that allows flow of blood back into the right atrium during systole	Fatigue, palpitations, dyspnea, abdominal bloating, anorexia, edema

Acute Rheumatic fever: treatment

	Specific Treatment Modalities	Rationale	Recommendations
Elimination of GAS skin or throat infection	Penicillin and alternatives	To limit the repetitive exposure to GAS antigens	A single intramuscular dose of BPG is ideal. Oral penicillin V can also be used. Azithromycin can be used for those with penicillin allergy.
Suppression of inflammation	NSAIDs (naproxen safer than aspirin) and steroids	Relief from symptoms of fever and arthritis NSAIDs and steroids have not been shown to prevent the progression to RHD	Naproxen immediate release oral: adult 250-500 mg q12h; child 10-20 mg/kg/d, divided q12h (maximum 1,000 mg/d) until pain is relieved, then taper the dose. Steroids are a useful alternative to NSAIDs in those who continue to worsen and develop heart failure. They enable rapid relief of symptoms and reduce the duration of hospitalization. Prednisolone/prednisone (1-2 mg/kg/d, maximum 80 mg/d) is generally preferred. It can be tapered after 2-3 weeks. Intravenous methylprednisolone can be used in life-threatening situations
Prevention of recurrences	Secondary penicillin prophylaxis	Secondary prophylaxis prevents recurrences of ARF and progression of heart valve damage	3-4 weekly intramuscular BPG or daily oral penicillin V (2009 AHA guidelines): ARF without carditis: until age 21 years or for 5 years after last ARF episode (whichever is longer) ARF with carditis but no residual valve damage: until age 21 years or for 10 years after last ARF episode (whichever is longer) ARF with carditis and residual valve damage: until age 40 years or for 10 years after last ARF episode (whichever is longer); lifelong prophylaxis may be needed

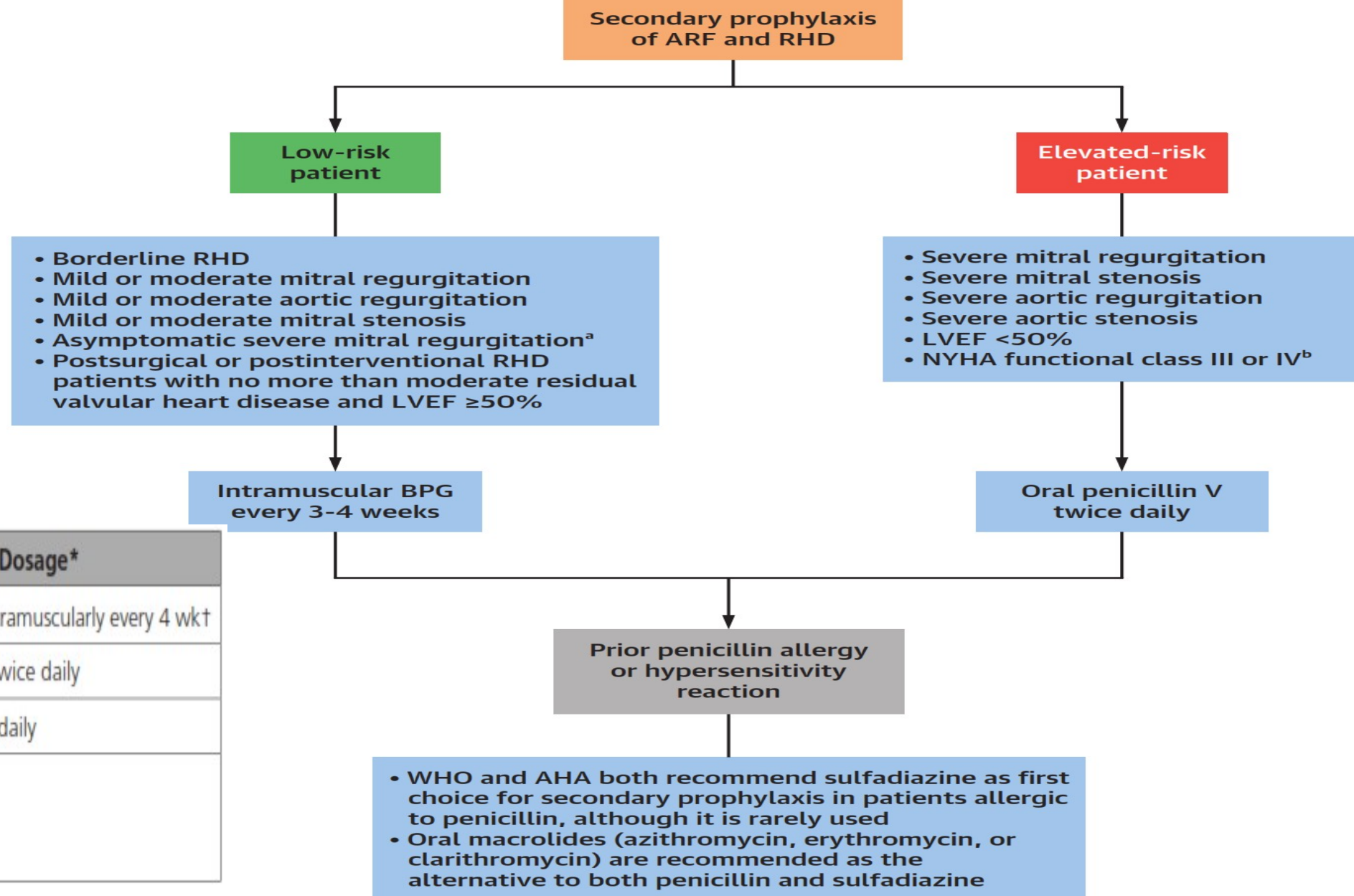
Acute Rheumatic fever: Treatment

Symptomatic management of arthritis/arthralgia after confirmation of ARF diagnosis	1. Naproxen immediate-release 250-500 mg (child 10-20 mg/kg/day) orally twice daily, up to a maximum of 1250 mg daily	Naproxen may be safer than aspirin, and convenient due to twice daily dosing and the capability oral suspension. Ibuprofen is well tolerated and readily available but data and experience with its use is less in ARF than naproxen. The dose of NSAIDS needed for ARF is generally higher than the dose recommended for other conditions, therefore it may be appropriate to commence at the higher dose range. Due to the rare possibility of Reye's syndrome in children, aspirin may need to be ceased during intercurrent acute viral illness, and influenza vaccination is strongly recommended.
	INDICATION	
	MEDICATION OPTIONS LISTED IN ORDER OF PREFERENCE	
	2. Ibuprofen 200-400 mg (child 5-10 mg/kg) orally three times daily, up to a maximum of 2400 mg daily	
	or	
	3. Aspirin adults and children 50-60 mg/kg/day orally, in four to five divided doses. Dose can be escalated up to a maximum of 80-100 mg/kg/day in four to five divided doses.	
Symptomatic management of moderate to severe chorea (Table 7.6)	1. Carbamazepine 3.5 to 10 mg/kg per dose orally, twice daily 2. Sodium valproate 7.5 to 10 mg/kg per dose orally, twice daily	Treatment of Sydenham chorea should be considered if movements interfere substantially with normal activities.
Symptomatic management of very severe chorea / chorea paralytica (Table 7.6)	In addition to an anticonvulsant agent, consider adding corticosteroid: Prednisone/prednisolone 1 to 2 mg/kg up to a maximum of 80 mg orally, once daily or in divided doses	

Acute Rheumatic fever: treatment

INDICATION	MEDICATION OPTIONS LISTED IN ORDER OF PREFERENCE	COMMENT
Symptomatic management of carditis	Paediatric dosing: Furosemide (frusemide) 1 to 2 mg/kg orally as a single dose, then 0.5 to 1 mg/kg (to a maximum of 6 mg/kg) orally, 6- to 24-hourly Spironolactone 1 to 3 mg/kg (initially) up to 100 mg orally, daily in 1 to 3 divided doses. Round dose to a multiple of 6.25 mg (a quarter of a 25 mg tablet). Enalapril 0.1 mg/kg orally, daily in 1 or 2 divided doses increased gradually over 2 weeks to a maximum of 1 mg/kg orally, daily in 1 or 2 divided doses, other ACE inhibitors (captopril, lisinopril, ramipril, perindopril)	Treatment of heart failure may be required in severe, acute carditis. Seek advice from a specialist cardiologist. Choice of ACE inhibitor will vary depending on the clinical situation. Seek advice from a specialist cardiologist.
	Digoxin 15 micrograms/kg orally, as a single dose, then 5 micrograms/kg after 6 hours, then 3–5 micrograms/kg (adult: 125–250 micrograms) orally, daily	Digoxin is rarely used in the treatment of acute carditis. Seek advice from a specialist cardiologist.
	Adult dosing: Furosemide (frusemide) 20–40 mg oral or intravenous as a single dose followed by 20–40 mg oral or intravenous 8–12 hourly. Ongoing dose adjustment based on clinical progression and renal function. Spironolactone may be added for patients having limited or no response to loop diuretic, 12.5–25 mg spironolactone orally daily Nitrate therapy may be added for patients having limited or no response to diuretic therapy, whose systolic blood pressure is greater than 90 mmHg. Intravenous or topical glyceryl trinitrate may be used. ACE inhibitor is recommended in patients with moderate or severe left ventricular systolic dysfunction, unless contraindicated	The management of acute carditis follows the same principles as the management of acute heart failure. This table gives a guide to the initial management of acute heart failure due to acute carditis in adults. Seeking advice from a specialist cardiologist early is strongly recommended.

Acute Rheumatic fever: treatment



Antibiotics for Prevention	Dosage*
Penicillin G benzathine	1.2 million U intramuscularly every 4 wkt
Penicillin V potassium	250 mg orally twice daily
Sulfadiazine	1 g orally once daily
Macrolide or azalide antibiotic (for patients allergic to penicillin and sulfadiazine)‡	Varies

AS: general management

Severe AS ($A < 1 \text{ cm}^2$)

- Avoid strenuous physical activity and competitive sports
- Avoid dehydration and hypovolemia
 - ✓ Significant reduction in cardiac output
- Drugs for hypertension or CAD
 - B-blockers, ACEIs/ARBs, diuretics, aldosterone receptor antagonists, and/or sacubitril/valsartan, SGLT2 inhibitors
- Vasodilators (e.g., sodium nitroprusside, nitroglycerin) in acute decompensated AS
- Nitroglycerin
 - For angina pectoris in patients with CAD
- Statin therapy for all patients with calcific AS
- Avoid diuretics in small LV because of the possible decrease in cardiac output
- Neither statins nor RAA-S- inhibitors prevent disease progression
- Endocarditis prophylaxis: if a prior history only
- Surgery
 - sAS and symptomatic, EF $< 50\%$
 - AS due to BAV disease and an aneurysmal root/ascending aorta

AR: treatment

Acute severe AR

- Intravenous diuretics and vasodilators (such as sodium nitroprusside)
- Beta blockers are best avoided;
 - ✓not to reduce the CO further or slow the heart rate
- **Surgery is** the treatment of choice, usually done within 24h of diagnosis

Chronic AR

• **Surgery**

- Onset of symptoms or LV systolic dysfunction

• **Medical therapy**

- ACEIs, ARBs, DHCCBs and hydralazine
 - ✓ Employed to treat HTN (SBP>140 mm Hg)
 - ✓ ACE inhibitors/ARBs are 1st line for HF from chronic AR, also sacubitril/valsartan is recommended in symptomatic severe AR and/or LV dysfunction when surgery is contraindicated.

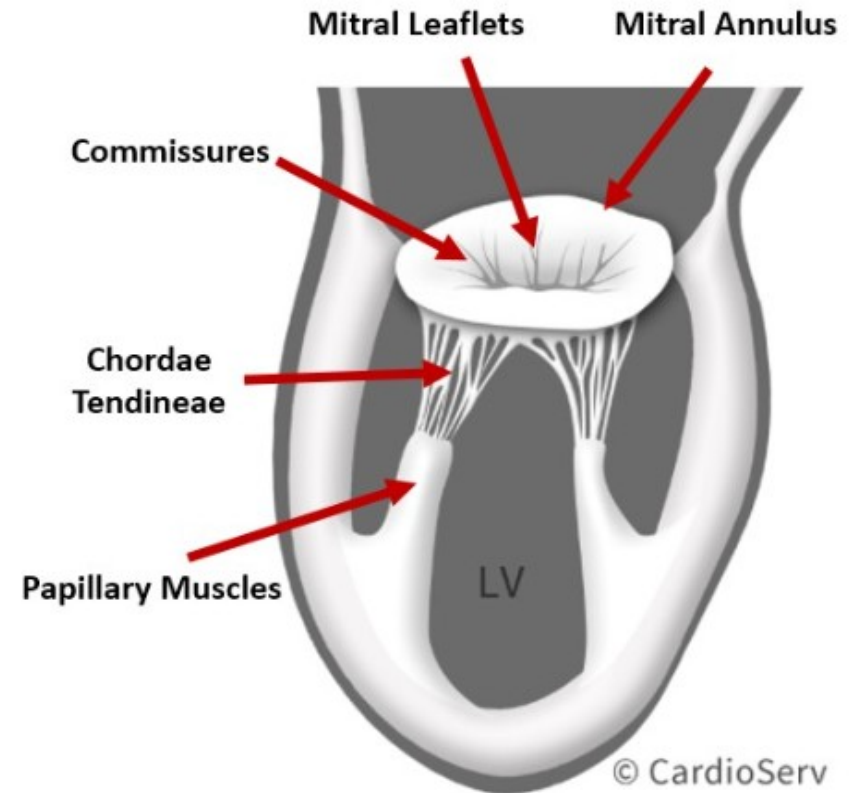
Mitral stenosis: treatment

- Secondary prophylaxis
 - Prevention of group A β -hemolytic streptococcal infection
- Symptomatic patients,
 - Restriction of sodium intake and small doses of oral diuretics
- Slowing HR in AF
 - Beta blockers, Non DHCCBs and digitalis
- **Warfarin** therapy targeting INR of 2–3 should be administered indefinitely
 - AF, a history of TE, or demonstrated LA thrombus
- **Recent onset AF** in non-severe MS not requiring PMBC or surgical intervention
 - Cardioversion
 - ✓ After 3 weeks of warfarin with therapeutic INR
 - Pharmacological with B-blocker/non-DHCCBs & digoxin

Mitral regurgitation

✍ Conditions affecting any of the five functional components of mitral valve apparatus;

- Leaflets, annulus, chordae tendineae, papillary muscles, and subjacent myocardium



Primary (Degenerative) MR: treatment

Chronic severe MR

- Anticoagulation

- Warfarin or a direct oral agent (e.g., apixaban, rivaroxaban)
 - ✓ If AF intervenes as guided by CHDA₂S₂-VASc risk score
- Avoid DOACs if moderate or severe rheumatic mitral stenosis and mechanical prosthetic valves

- Cardioversion depending up on;
 - Clinical context, AF chronicity, and LA size
- Chronic, isolated severe MR with preserved LV systolic function and systemic HTN
 - Vasodilators therapy

MR and CHDA₂S₂-VASc

Risk factor	Score
Congestive heart failure/ LV dysfunction	1
Hypertension	1
Age ≥ 75 years	2
Diabetes mellitus	1
Stroke/TIA/thromboembolism	2
Vascular disease (prior MI, peripheral artery disease, aortic plaque)	1
Age 65-74 years	1
Sex category (female gender)	1
Maximum possible score (since age may contribute 0,1 or 2 points)	9

Secondary (functional) MR: treatment

- Severe secondary MR with ischemic or dilated cardiomyopathy
 - Aggressive GDMT of heart failure when indicated
 - ✓ Diuretics, β Bs, ACEIs/ARBs, ARNI, mineralocorticoid receptor antagonists, SGLT2IS and biventricular pacing (cardiac resynchronization therapy)
 - Antibiotics prophylaxis
 - Prior history of IE
- ✍ Acute severe MR
 - Urgent stabilization and preparation for surgery
 - Diuretics, intravenous vasodilators (particularly sodium nitroprusside), and even mechanical support
 - ✓ Post-MI papillary muscle rupture or other forms of acute severe MR

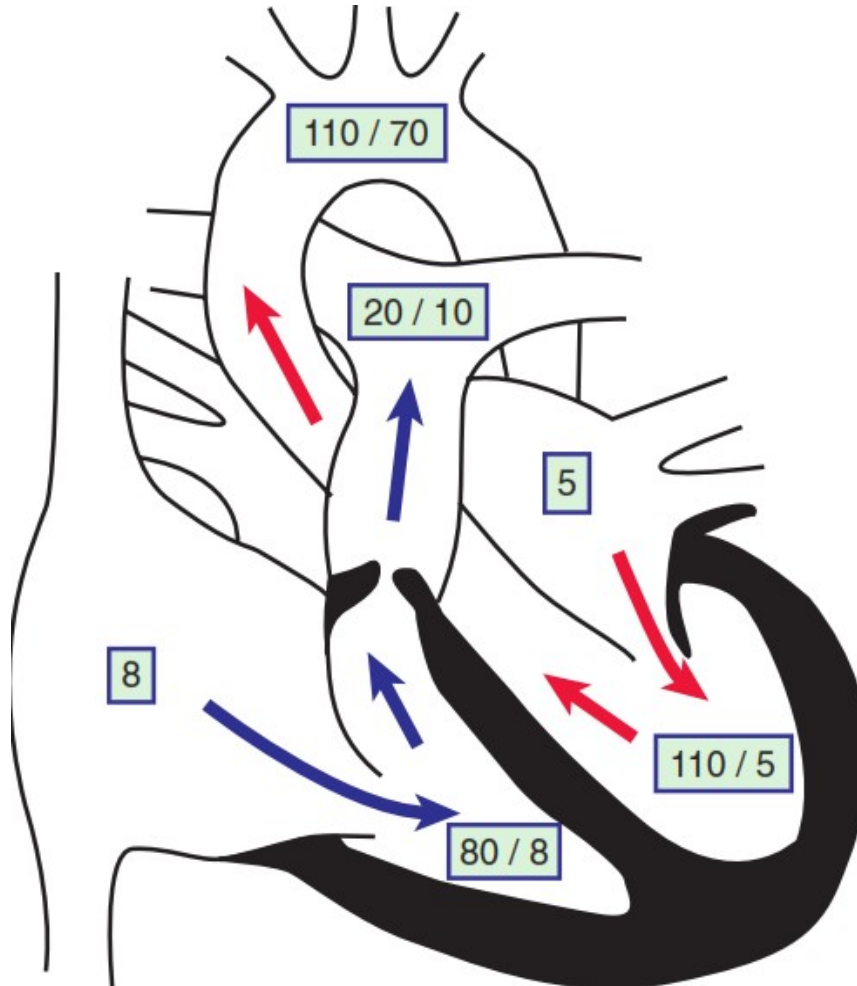
TR: treatment

- Severe TR and signs of right HF
 - Diuretics
- An aldosterone antagonist
 - As many have secondary hyperaldosteronism from marked hepatic congestion
- If PA hypertension and severe secondary TR;
 - Therapies to reduce PH and/or pulmonary vascular resistance, including those targeting left-sided heart disease

Tricuspid valve surgery

- Moderate to severe TR
 1. Undergoing left-sided valve surgery
 2. Tricuspid annular dilation (>40 mm)
 3. History of right heart failure
 4. PA hypertension
- Repair than replacement

Pulmonic stenosis



- Diuretics to treat symptoms and signs of right heart failure
- Percutaneous pulmonic balloon valvuloplasty
 - Symptomatic patients with moderate or severe PS and asymptomatic patients with peak gradient >64 mmHg

Pulmonic regurgitation

- Mild or moderate PR do not, by themselves, result in symptoms
- PA hypertension, may dominant the clinical picture
- With progressively severe PR and RV dysfunction,
 - Fatigue, exertional dyspnea, abdominal fullness/bloating, and lower extremity swelling
- Functional PR due to PA hypertension and annular dilation
 - Vasodilators and/or surgical/interventional, based on cause of PH (e.g., idiopathic PA hypertension, left-sided heart valve disease)
- Right heart failure symptoms
 - Diuretics
- Surgical valve replacement
 - Primary, severe, pulmonic valve disease, such as carcinoid or endocarditis, is rarely undertaken.

Mixed Valvular Heart disease

- Management can be challenging
 - Follow the dominant lesion
- Limited medical therapies
 - Diuretics if congestion and anticoagulation to prevent stroke and thromboembolism in AF
- Blood pressure-lowering medications
 - If systemic hypertension which may aggravate left-sided regurgitant valve lesions
- Pulmonary vasodilators to lower PVR are not generally effective in this context
- There is a paucity of evidence;
 - Surgical and/or transcatheter valve intervention in patients with multiple or mixed valve disease
- But when there is a clear, dominant lesion
 - Such as severe AS and mild AR follow the recommended for intervention in patients with AS